Huawei Access Node - Routing Protocol Flap (BGP, OSPF, ISIS) Failure Resolution SOP

Node: BAS-ACC-05 | **Vendor:** Huawei | **Severity:** Critical | **Alarm Code:** ALARM_3557

Description

This Standard Operating Procedure (SOP) outlines the steps to be taken in the event of a Routing Protocol Flap (BGP route churn, OSPF adjacency loss, ISIS LSP flooding) on Huawei Access Node BAS-ACC-05 located at Basavanagudi POP. The failure occurs due to the complex interplay of BGP, OSPF, and ISIS protocols, which are used to establish and maintain routing table information across the network. The protocol flapping can be caused by a variety of factors, including misconfiguration of routing protocols, hardware failures, or software issues. In the case of Huawei Access Nodes, the issue is further complicated by the use of advanced features such as MPLS and VPNs, which can lead to unpredictable behavior in the event of a protocol flap.

Pre-requisite Checks

- check: Physical Site Visit, description: Ensure that a physical visit to the Basavanagudi POP is made to verify the issue and perform any necessary troubleshooting steps.

- check: Redundant Link Status, description: Verify the status of the redundant links to ensure that they are functioning correctly and not contributing to the protocol flap.

- check: Power and Cooling, description: Verify that the power and cooling systems at the Basavanagudi POP are functioning correctly and are not contributing to the issue.

Potential Root Causes

Cause	Probability
Misconfigured Routing Protocols	High
Hardware Failure	Medium
Software Issues	Low

Diagnostic Commands & Expected Outputs

```
Command: display bgp ipv4 unicast summary, Expected Output: BGP summary information, Description: This command will display the BGP summary information, including the number of established peer connections, the number of received and sent updates, and the number of routing table entries.

Command: display ospf ipv4 routing-table, Expected Output: OSPF routing table information, Description: This command will display the OSPF routing table information, including the routing table entries, the next-hop IP address, and the metric value.
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```
Command: display isis database, Expected Output: ISIS database information, Description: This command will display the ISIS database information, including the LSP (Link-State Packet) entries, the LSP sequence number, and the LSP checksum.
```

Resolution Steps

Step 1: Step: Verify Routing Protocol Configuration, Command: display bgp ipv4 unicast conf, Description: Verify that the BGP configuration is correct and that there are no misconfigurations that could be causing the protocol flap.

Step 2: Step: Verify OSPF Configuration, Command: display ospf ipv4 conf, Description: Verify that the OSPF configuration is correct and that there are no misconfigurations that could be causing the protocol flap.

Step 3: Step: Verify ISIS Configuration, Command: display isis conf, Description: Verify that the ISIS configuration is correct and that there are no misconfigurations that could be causing the protocol flap.

Step 4: Step: Clear BGP Cache, Command: clear bgp ipv4 unicast cache, Description: Clear the BGP cache to remove any stale routing information that may be contributing to the protocol flap.

Step 5: Step: Clear OSPF Cache, Command: clear ospf ipv4 cache, Description: Clear the OSPF cache to remove any stale routing information that may be contributing to the protocol flap.

Step 6: Step: Clear ISIS Cache, Command: clear isis cache, Description: Clear the ISIS cache to remove any stale routing information that may be contributing to the protocol flap.

Post-Check Verification

```
Command: display bgp ipv4 unicast summary, Expected Output: BGP summary information, Description: Verify that the BGP summary information is correct and that the protocol flap has been resolved.

Command: display ospf ipv4 routing-table, Expected Output: OSPF routing table information, Description: Verify that the OSPF routing table information is correct and that the protocol flap has been resolved.

Command: display isis database, Expected Output: ISIS database information, Description: Verify that the ISIS database information is correct and that the protocol flap has been resolved.
```

Roll-back Procedures

- Step: Restore Original Configuration, Command: undo bgp ipv4 unicast conf, Description: Restore the original BGP configuration to resolve the issue.

- Step: Restore Original OSPF Configuration, Command: undo ospf ipv4 conf, Description: Restore the original OSPF configuration to resolve the issue.

- Step: Restore Original ISIS Configuration, Command: undo isis conf, Description: Restore the original ISIS configuration to resolve the issue.